

Comparative Structured Observation

A Mixed Qualitative Method for “Getting the Design Right”

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Abstract

What happens when your design concept is already well established and you want to see whether or not your design is on the right track, or which of several design variants makes most sense to users? You are not interested in testing a hypothesis, but instead, you want gather focused, useful feedback from users that either supports your current design direction or helps you decide what to change. This course presents a qualitative method called *Comparative Structured Observation (CSO)* (Mackay & McGrenere, 2025) which takes advantage of the structure of controlled experiments to generate comparable, ecologically relevant experiences with two or more design variants. Course participants will first learn the basic principles that comprise a CSO study, with examples from the research literature, and a discussion of what constitutes appropriate and inappropriate study designs. Participants will then work in small groups and, with help from the instructors, to design concrete CSO studies. Participants are encouraged to bring examples of current design concepts that they are struggling to assess, but will also have other examples to choose from. The course will conclude with a discussion of relevant analysis methods

CCS Concepts

• **Human-centered computing** → **HCI design and evaluation methods**.

Keywords

Interaction Design Methods, Mixed Methods, User Experience Design, Qualitative Methods

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1 Benefits

The goal of this course is to provide participants with an overview of the Comparative Structured Observation method for assessing

a current design concept by gathering participants’ reflections as they compare different design variants of a design concept. Participants will first learn the basics of the method, with examples from the HCI research literature. Participants will then gain hands-on experience as they work in small groups to design a CSO study that assesses a specific design concept or distinguishes among design variants. Participants may choose from examples provided by the instructors or bring their own hard-to-assess design concepts or variants. Finally, participants will learn how to efficiently analyze the resulting qualitative data.

2 Intended audience(s)

The primary audience for this course is HCI Researchers, often Ph.D. students, who have developed an interactive system in the form of a low- or medium fidelity prototype and now need to assess the design concept or choose among possible design variants. UX designers who are developing novel product designs will also benefit.

3 Prerequisites

The course does not include any prerequisites, beyond an interest in assessing interactive technologies that meet the needs of real users in real-world contexts. However, participants should have a basic knowledge of experiment design.

4 Content

The course is based on the paper: *Comparative Structured Observation* [1] to appear in ACM/TOCHI, which will be presented at ACM/CHI’25. The primary goal of Comparative Structured Observation is to capture insights from participants’ reflections about selected design variants in order to generate implications for design, either by identifying the aspects of the design that do or do not work well and why, or by providing direction for how to improve the design. Comparison is essential — both participants and researchers must be able to compare and reflect on the advantages and disadvantages of each variant relative to its context of use. Because Comparative Structured Observation often yields insights about how different tasks moderate the success of different design choices, sometimes characterized as What design works When? How? and Why?, this approach offers richly contextualized implications for design.

The introductory lecture will first define and characterize the Comparative Structured Observation method as:

an interventionist, qualitative method for assessing and advancing a design concept where researchers observe participants as they compare and reflect deeply

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Dimension	#	Study qualifies as a Comparative Structured Observation if it ...	Study qualifies as a good Comparative Structured Observation if it ...
Design Concept Basis	1	builds on design concepts influenced by formative research, ideally conducted by the researcher but also from the literature.	reports on substantive formative research, e.g. a well-run participatory design workshop that includes reflection by both participants and researchers.
Role of Comparison	2	ensures that each participant experiences at least two design variants in the study, e.g. different novel designs, variants within a novel design, or a baseline.	chooses design variants that meaningfully advance the design concept(s) and avoids straw-man comparisons.
	3	structures participant activities so they can experience and compare design variants, e.g. perform equivalent tasks with each design variant.	chooses and structures meaningful activities for participants, e.g., ecologically valid tasks.
	4	structures comparisons according to experimental design or quasi-experimental design principles, e.g. counter-balance tasks for order.	justifies the protocol relative to the setting (lab or field) and comparisons being made, according to best experimental or quasi-experimental design practices.
Type of Data Collected	5	records participants' comparisons and reflections on the qualitative differences in their experiences with the design variants, e.g., through interview questions.	includes well-designed interviews or surveys that elicit detailed, thoughtful comparisons by participants after exposure to the design variants.
	6	records participants' interactions with each design variant, e.g. through video recordings or high-quality cinematic logs.	collects rich, in-situ observational data or the best-possible alternative, e.g. remote video or substantive experience samples.
	7	records quantitative data only if it helps add context to qualitative data; e.g. percentage of time participants spent in an activity.	records quantitative data, if relevant, to contextualize qualitative data, e.g. participants' interactions with design elements that clarify their experiences.
Type of Data Analysis	8	analyzes participants' comparisons and reflections about the design variants; e.g. with reflexive thematic analysis.	demonstrates that participants have compared and reflected deeply about their experiences with the design variants.
	9	analyzes researchers' independent assessment of the participants' experiences; e.g. with reflexive thematic analysis.	leverages rich, qualitative data so that researchers can independently assess participants' reflections.
	10	treats qualitative analysis as primary.	conducts and reports a rigorous qualitative analysis according to the best practices of a well-established qualitative method.
	11	treats quantitative analysis as secondary.	analyzes quantitative data according to the best practices of well-established quantitative methods, either or both descriptive or inferential statistics.
Results	12	reports findings and analysis to advance one or more design concept(s).	explicitly discusses the implications for design and how the design concept(s) should evolve, based on the study results.

Table 1: Identifying the key characteristics of a *Comparative Structured Observation* study (left column), and considerations for what makes a “good” study (right column).

upon their experiences with selected design variants, exposure to which is structured according to experimental design principles.

We will explain the rationale for why the method requires comparisons, structure and observation, and will provide a brief history of examples of how the method has been used successfully in the HCI literature. We will then explain the specific characteristics of a CSO study (Figure 1) and identify the characteristics of a “good” one.

5 Practical work

After learning the basic method, course participants will work in groups to design their own CSO studies. Although instructors will provide examples of possible design concepts, participants are encouraged to bring their own concepts or design variants and explain which particular aspects they would like to assess. The instructors will work with each group to ensure that they create

effective CSO study designs and will also explain the best approach for analyzing the resulting qualitative data.

6 Instructor background

Wendy E. Mackay is a Research Director, Classe Exceptionnelle at Inria, the National French Research Organization for Computer Science, the 2021-2022 Annual Chair for Computer Science at the Collège de France and recipient of the CHI Lifetime Research award. She holds a Ph.D. from MIT, as well as degrees in Computer Science and Experimental Psychology. She has been teaching about HCI, specifically design methodology, at the Université Paris-Saclay for two decades, as well as at Stanford University, Aarhus University, at CHI'23 and for various corporate audiences.

Joanna M. McGrenere is a Professor of Computer Science and is Co-Chair of the Computer Science Department at the University of British Columbia, Canada. She holds a Ph.D. from the University of

Toronto and is an ACM Distinguished Member. She teaches about HCI at both the undergraduate and graduate level, with a specific research interest in adaptive user interfaces and universal usability.

Both instructors have taught this material to their respective HCI research groups, as well as part of Master's and Ph.D. level HCI methodology courses. The course has also been taught in Switzerland as part of an HCI local special interest group, with 80 participants, although that version did not include the hands-on exercises. The instructors have also been invited to run additional courses and workshops about the method after the CHI'25 conference.

7 Resources

The course is based on the soon-to-appear ACM/TOCHI paper *Comparative Structured Observation (CSO)* [1], which will also be presented at the CHI'25 conference. During the hands-on portion of the course, participants will need access to flipchart-sized paper, sticky notes and pens. Participants will have access to a website with copies of the paper, slide handouts and course notes.

8 Accessibility

The course is fully accessible for people with motor impairments. Participants with either visual or auditory impairments will be able to participate but we do not have sign language interpretation or non-visual course materials.

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References

- [1] Wendy E. Mackay and Joanna McGrenere. 2025. Comparative Structured Observation. *Trans. Human-Computer Interaction* (2025). In print.